

CHANDIGARH UNIVERSITY APEX INSTITUTE OF TECHNOLOGY



EXPERIMENT 7

**Subject: Advanced Database Management
System**

**Name: Rishav Giri
Class:24AIT-KRG-1-B
UID:24BAI70369**

Submitted To: Alok Sir

**1. Consider a relation R having attributes as R(ABCD), functional dependencies are given below:
AB→C, C→D, D→A**

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Ans:

Closure:

B+= B
BA+= BACD
BC+= BCDA
BD+= BDAC

- Prime Attributes = {A,B,C,D}
- Super set / Candidate set = {AB, BC, BD}

Since, all the RHS attributes are prime, the relation R is in 3NF.

Check for BCNF:

For BCNF, for every non-trivial FD $X \rightarrow Y$, X must be a superkey.

$C \rightarrow D$, C is not a superkey, therefore it violates BCNF.

Highest Normal Form: 3NF

2. Relation R(ABCDE) having functional dependencies as :

A→D, B→A, BC→D, AC→BE

Identify the set of candidate keys possible in relation R. Find the highest normal form.

Ans:

Closure:

C+=C
CA+=CADBE
CB+=CBAED
CD+=CD
CE+=CE

- Candidate sets = {CA, CB}
- Prime attributes = {A, B, C}

Since in $A \rightarrow D$, D can be determined by only A (subset of candidate key AC), it violates partial dependency so it is not in 2NF.

Highest Normal Form: 1NF

3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:

B→A, A→C, BC→D, AC→BE

Identify the set of candidate keys possible in relation R. Find the highest normal form.

Ans:

Closure:

B+=BACDE
A+=ACBED
C+=C
E+=E

$$D^+=D$$

- Candidate keys: {A, B}
- Primary keys: {A, B}

Since all the attributes on left side are either candidate keys or super keys, hence the relation R is in BCNF form.

Highest Normal Form: BCNF

**4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:
 $A \rightarrow BCD$, $BC \rightarrow DE$, $B \rightarrow D$, $D \rightarrow A$
 Find the highest normal form.**

Ans:

F never appears on the RHS of any FD, therefore it must be present in every candidate key.

$$\begin{aligned} AF^+ &= AFBCDE \\ BF^+ &= BFDACE \\ CF^+ &= CF \\ DF^+ &= DFABCE \\ EF^+ &= EF \end{aligned}$$

- Candidate keys: {AF, BF, DF}
- Prime attributes: {A, F, B, D}

$A \rightarrow C$

A is a proper subset of candidate key AF, and C is non-prime.
 Therefore, there is a partial dependency, so the relation violates 2NF.

Highest Normal Form: 1NF

5. Relation R(ABCDE) having dependencies as:

**$CE \rightarrow D$,
 $D \rightarrow B$,
 $C \rightarrow A$**

Highest normal form? C.K set?

Ans:

Closure:

$$CE^+ = CEDBA$$

- Candidate key = {CE}
- Prime keys = {C, E}

From FD, $C \rightarrow A$

C is a proper subset of candidate key CE, and A is non-prime.
 Therefore, $C \rightarrow A$ is a partial dependency which violates 2NF.

Highest Normal Form: 1NF

6. Relation R(ABCDEF) having dependencies as:

AB → C

DC → AE

E → F

Highest normal form? C.K set?

Ans:

Since B, D never occur on RHS of any FD, they will be a part of candidate key.

Closure:

BD+=BD
BDA+=BDACEF
BDC+=BDCAEF
BDE+=BDEF
BDF+=BDF

- Candidate keys= {BDA, BDC}
- Prime attributes= {B, D, A, C}

Check for 2NF:

- DC → AE: A is prime but E is non-prime

Here, DC is a proper subset of candidate key BDC.

Therefore DC → E is a partial dependency so 2NF is violated.

Highest Normal Form: 1NF

7. Relation R(ABCDEFGHI) having dependencies as:

AB → C

BD → EF

AD → GH

A → I

Highest normal form? C.K set?

Ans:

Since A, B, D never occur on RHS of any FD, they will be a part of candidate keys.

Closure:

ABD+=ABDCEFGHI

- Candidate set = {ABD}
- Primary attributes = {A, B, D}

Check for 2NF:

- AB → C: AB is a proper subset of ABD and C is non-prime. Therefore:

AB → C is a partial dependency, so the relation violates 2NF.

Highest Normal Form: 1NF

8. Relation R(ABCDE) having dependencies as:

AB → CD

D → A

BC → DE

Highest normal form? C.K set?

Ans:

Since B doesn't appear on RHS, it must be the part of every candidate key.
Closure:

BC+=BCDEA
AB+=ABCDE
BD+=BDACE
BE+=BE

- Candidate key={BC, BA, BD}
- Prime attributes= {B, C, A, D}

Check for 3NF:

- **AB -> CD**: AB is a super key
- **D -> A**: D is not a super key but A is a prime attribute
- **BC -> DE**: BC is a super key

Check for BCNF:

- **D -> A**: D is not a super key

Therefore it violates BCNF

Highest Normal Form: 3NF

9. Relation R(ABCDE) having dependencies as:

BC -> ADE

D -> B

Highest normal form? C.K set?

Ans:

Since C does not appear on RHS, it must be a part of candidate key
Closure:

C+=C
BC+=ADEBC
CD+=CDBAE
CE+=CE

- Candidate keys={BC, CD}
- Prime attributes={B,C,D}

Check for 3NF:

- **D -> B**: D is not a super key, but B is a prime attribute
- **BC -> ADE**: BC is a super key

Check for BCNF:

- **D -> B**: D is not a super key, it violates BCNF

Highest Normal Form: 3NF

10. Relation R(ABC) having dependencies as:

A → B

B → C

C → A

Highest normal form? C.K set?

Ans:

Closure;

A⁺=ABC

B⁺=ABC

C⁺=ABC

- Candidate keys={A,B,C}
- Prime attributes={A,B,C}

Since all the LHS attributes are super keys,

Highest Normal Form= BCNF

11. Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

F1 → F3

F2 → F4

(F1,F2) → F5

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Ans:

Since F1, F2 doesn't appear on RHS, they must be a part of candidate key.

Closure:

F1F2⁺=F1,F2,F3,F4,F5

- Candidate set={F1F2}
- Prime attributes= {F1, F2}

Check for 2NF:

- F1 → F3: F1 is a subset of candidate key F1F2 and F3 is a non-prime attribute, therefore it violates 2NF
- F2 → F4: F2 is a subset of candidate key F1F2 and F4 is a non-prime attribute, therefore it violates 2NF

Highest Normal Form: 1NF